

Comparative genomics

Signs of selection

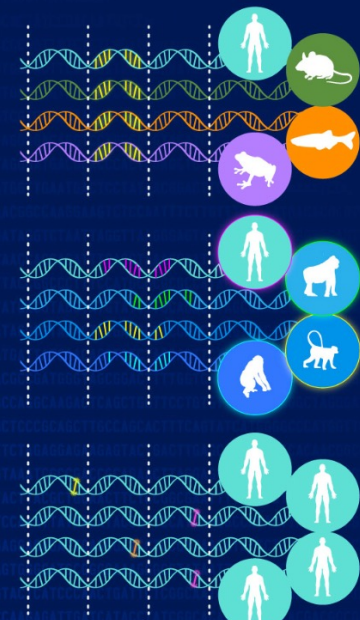
Bogotá,
Colombia
2026



Comparative genomics

- **Comparative genomics** is the field of biology that compares the genome sequences of different organisms to understand their evolutionary relationships and biological function.
- Comparing genomes
 - from within individual, sister species, all the way to between very divergent taxa.

Comparative Genomics



NHGRI FACT SHEETS
genome.gov

Researchers choose the appropriate time-scale of evolutionary conservation for the question being addressed.

Common features of different organisms such as humans and fish are often encoded within the DNA evolutionarily conserved between them.

Looking at closely related species such as humans and chimpanzees shows which genomic elements are unique to each.

Genetic differences within one species such as our own can reveal variants with a role in disease.

NIH National Human Genome Research Institute

Comparative genomics

- **Studying genome evolution itself** — how chromosomes rearrange, how gene families expand or contract, how much of a genome is repetitive DNA, etc.
- **Identifying functional elements** — sequences conserved across many species are often functionally important (genes, regulatory regions)
- **Reconstructing evolutionary history** — building phylogenetic trees, dating when species diverged, tracing genome changes through time
- **Understanding gene function** — if a gene's function is known in one species (e.g., a well-studied model organism like mouse or yeast), finding counterpart (an *ortholog*) in another species
- **Applied uses** — crop and livestock breeding, tracking pathogen evolution, conservation genetics, local adaptation and human disease research

Comparative genomics using assembled genomes

- Genome assembly
 - Reconstruction of the genome of one or more individuals
- Reference genome
 - Genome assembly that is the point representation of the structure and organisation of the genome of a species



Genome

Download a genome data package including genome, transcript and protein sequence, annotation and a data report

Selected taxa

Homo sapiens Enter one or more taxonomic names

Filters

Download Select columns 1,896 Genomes Rows per page 20 1-20 of 1,896

☐ Assembly	GenBank	RefSeq	Scientific name	Modifier	Annotation	Action
☐ GRCh38.p14	GCA_000001405.29	GCF_000001405.40	Homo sapiens (human)		NCBI RefSeq	
☐ T2T-CHM13v2.0	GCA_009914755.4	GCF_009914755.1	Homo sapiens (human)		NCBI RefSeq	
☐ hg002v1.1.mat	GCA_018852615.3		Homo sapiens (human)	NA24385 (isolate)		
☐ hg002v1.1.pat	GCA_018852605.3		Homo sapiens (human)	NA24385 (isolate)		
☐ PAN027_pat_v1.0	GCA_046332035.1		Homo sapiens (human)	HG06807 (isolate)		
☐ RPE1V1.1 Haplotype 1	GCA_050656345.1		Homo sapiens (human)	RPE-1 (isolate)		

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National Library of Medicine
National Center for Biotechnology Information

Genome assembly GRCh38.p14

reference

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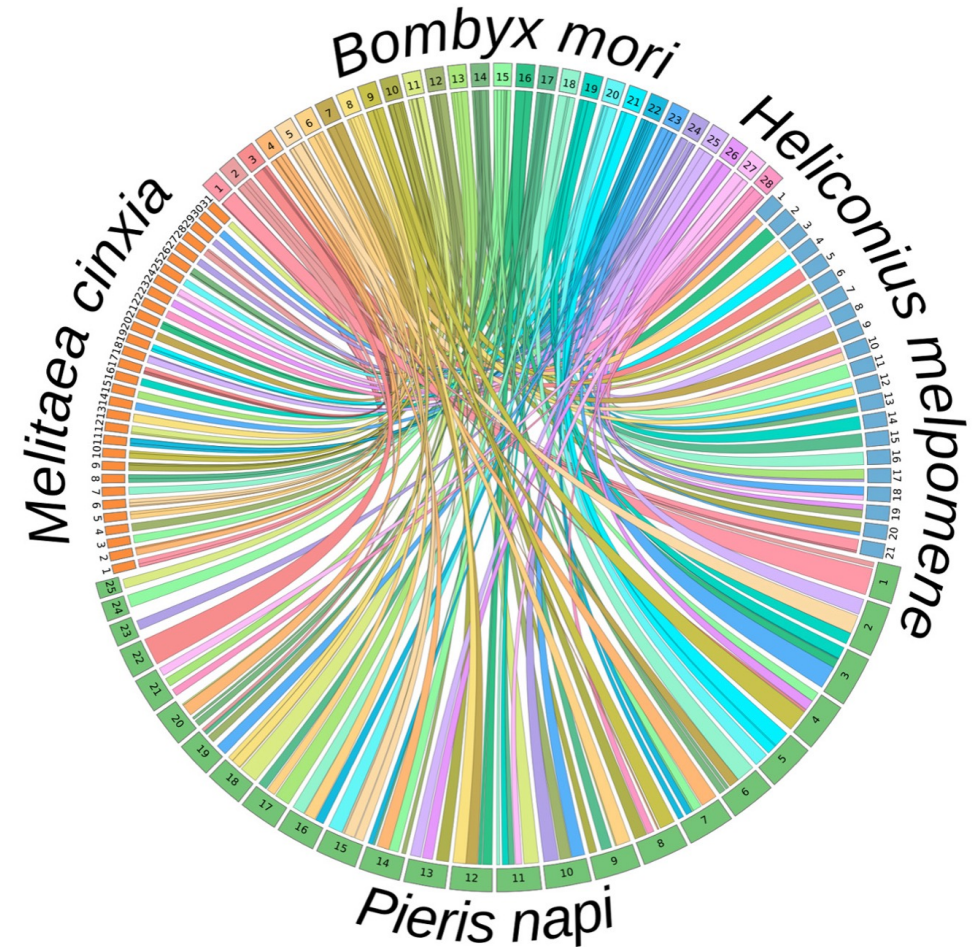
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Comparative genomics

- Detect signs of selection using dN/dS
- Genome structure - size
- Genome features
 - Gene and regulatory dynamics
 - Gene family dynamics
 - Conserved non-coding elements
 - TEs and repeats
 - CG-content, codon usage and tRNA dynamics
- Heterozygosity, RoH
- Genome structure - karyotype

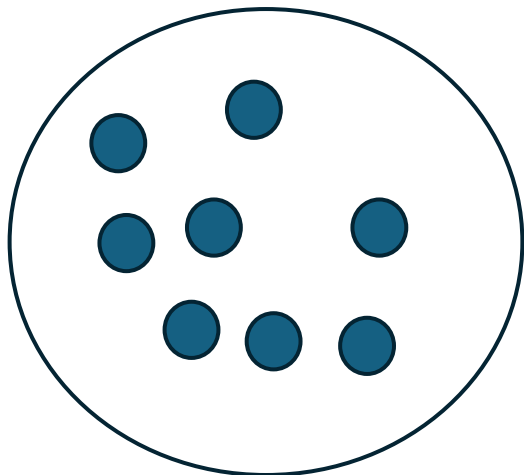
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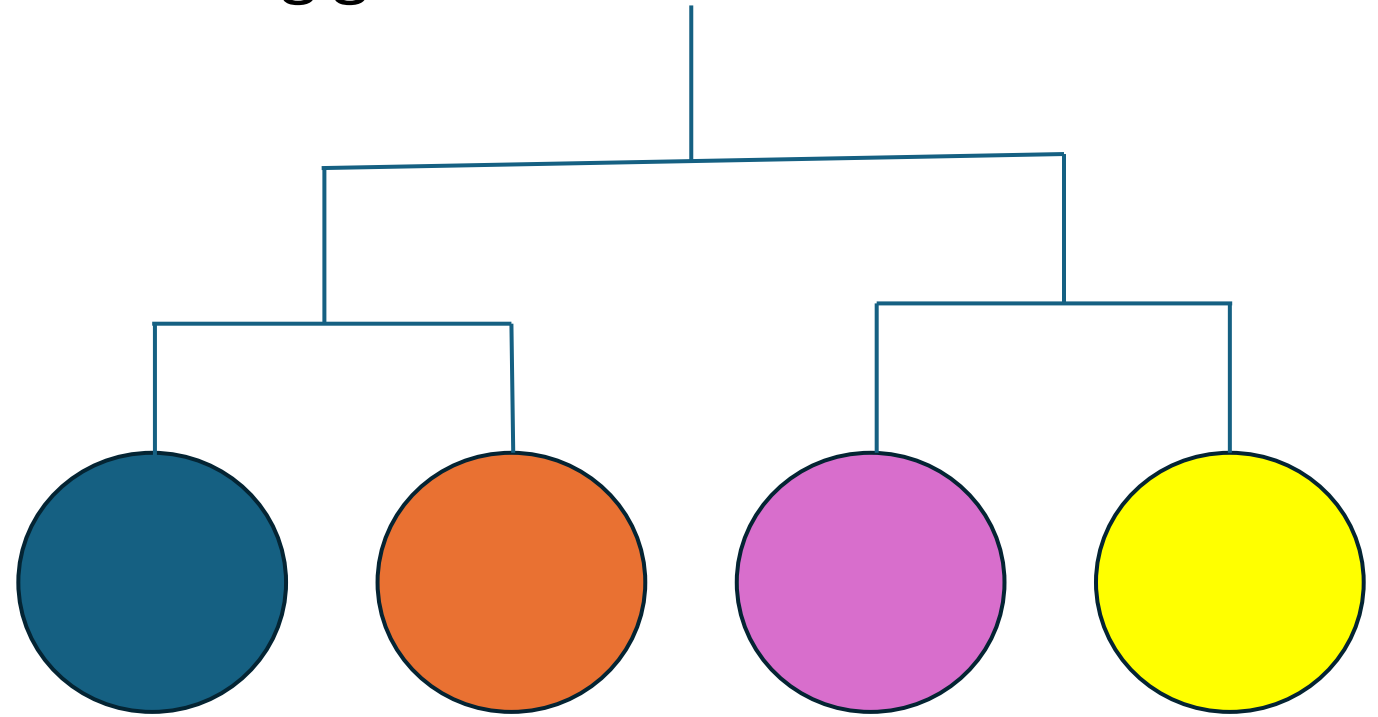
Hill et al. 2019

Detect signs of selection using dN/dS

Moving away from population data to larger evolutionary distances to detect genes under selection using genome assemblies from different species



Population



Species 1

Species 2

Species 3

Species 4

Genes under selection

- Most mutations are neutral or slightly deleterious
- Most genes are under purifying selection
- Beneficial mutations are rare
 - Change in environment - local adaption
 - Novel pathogens or predators
 - Gene duplication events – release from purifying selection



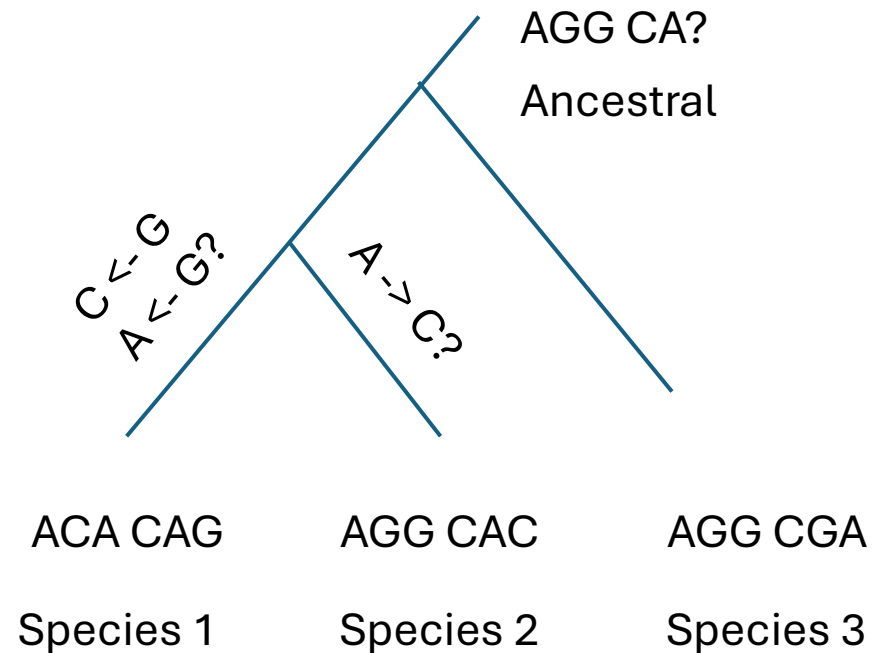
Photo by [Nikolai Vladimirov via iNaturalist](#), [CC BY-NC 4.0](#))

Single nucleotide mutations in a protein in the voltage-sensitive sodium channel (para). Confers nerve insensitivity to DDT and pyrethroid insecticides – under strong positive selection.

Martinez-Torres et al., 1998, Williamson Mol Gen Genet (1993) 240:17-22

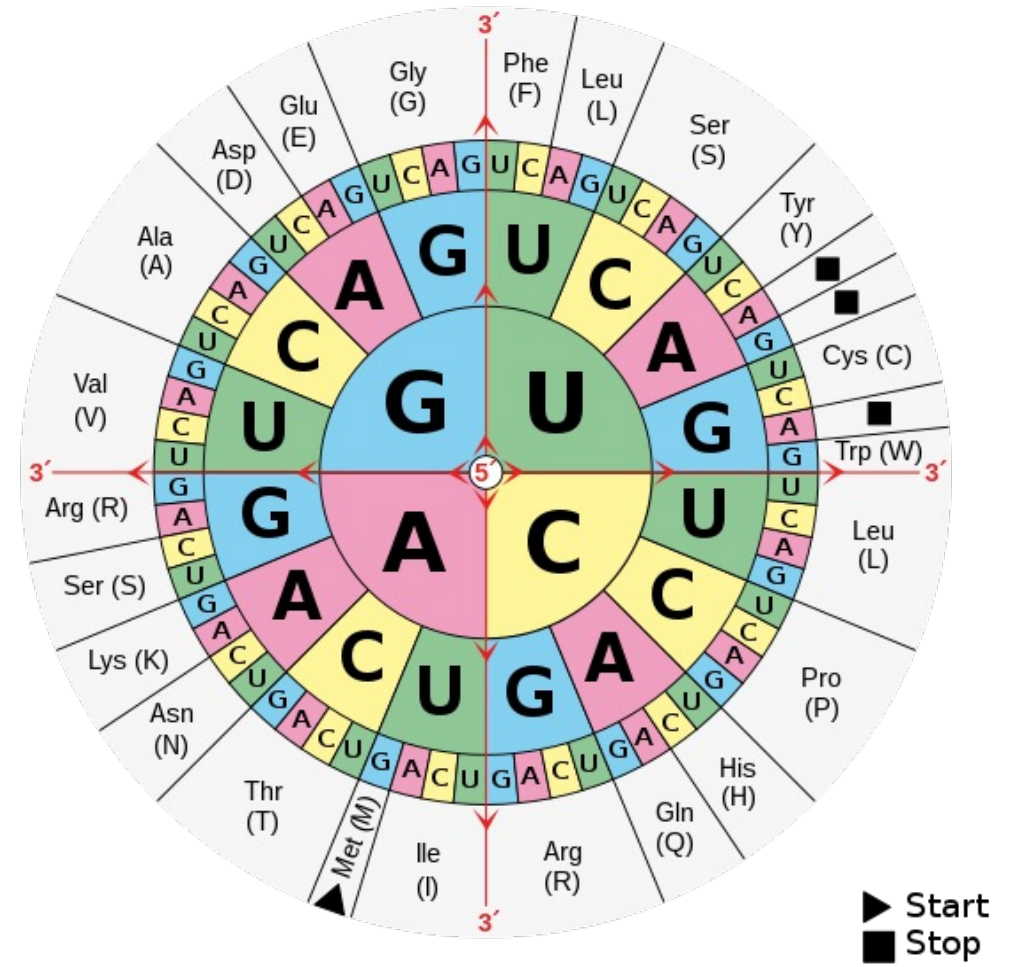
How to detect genes under selection using genome assemblies

- Substitution-based detection in a phylogenetic framework
 - **Substitutions** are single nucleotide **mutations** that have been fixed in a species
- Ancestral state reconstruction
- Substitution rates along branches



Substitution types - nomenclature

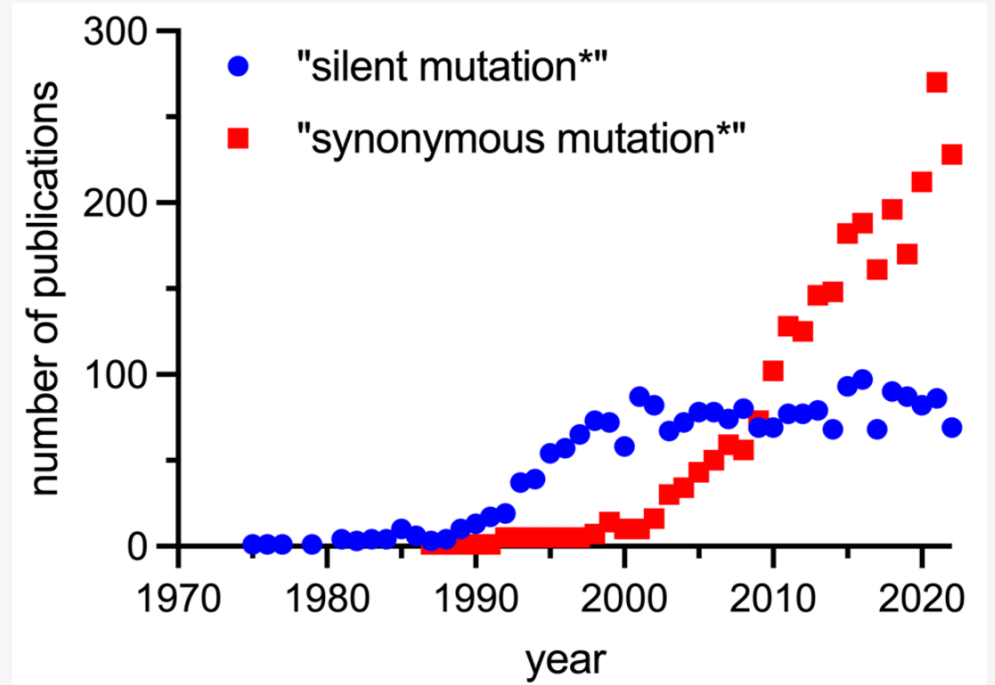
- Synonymous substitutions – no change in amino acid
- Non-synonymous substitutions
 - Change in amino acid - missense
 - From amino acid to stop codon - nonsense
 - From stop codon to amino acid - nonstop



Substitution types - nomenclature

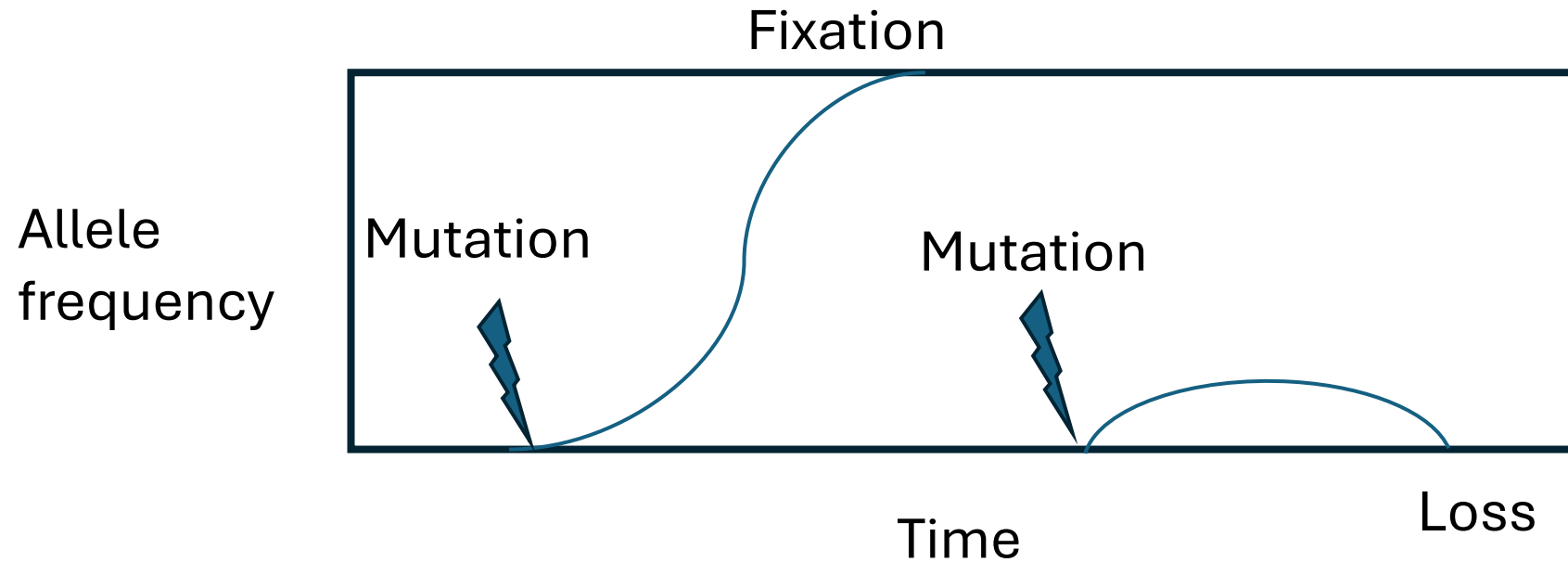
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Figure 3. Number of publications found in PubMed using the search terms "silent mutation*" and "synonymous mutation*".



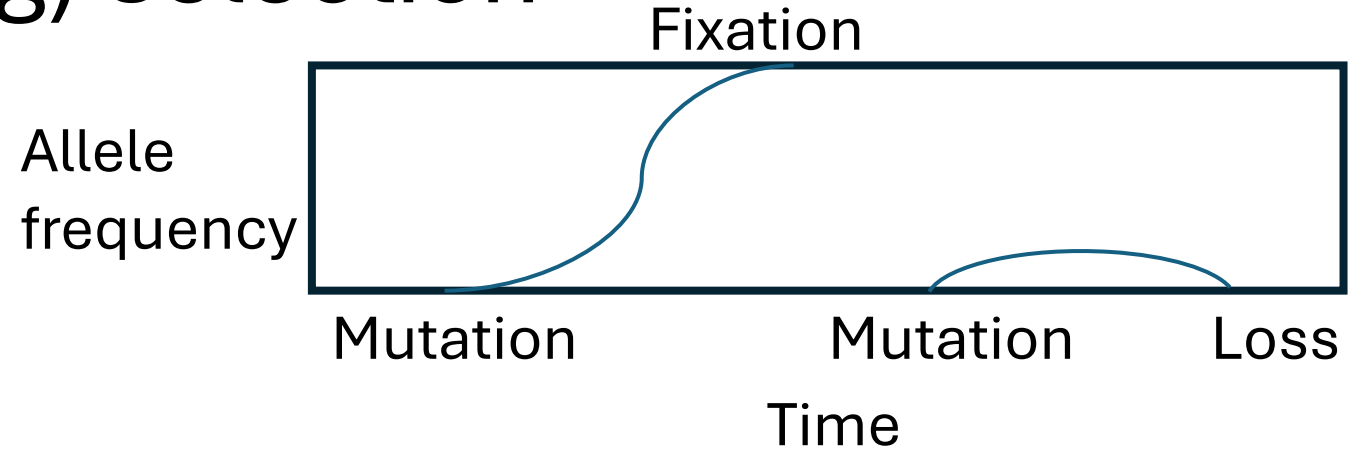
Substitutions

In neutral positions the loss or fixation of a mutations is random (synonymous substitutions)

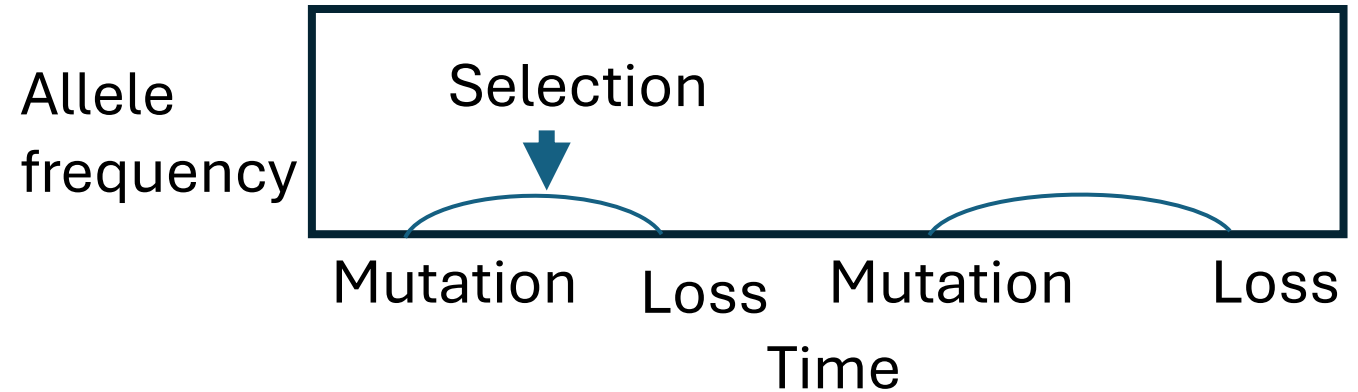


Negative (purifying) selection

Synonymous substitutions

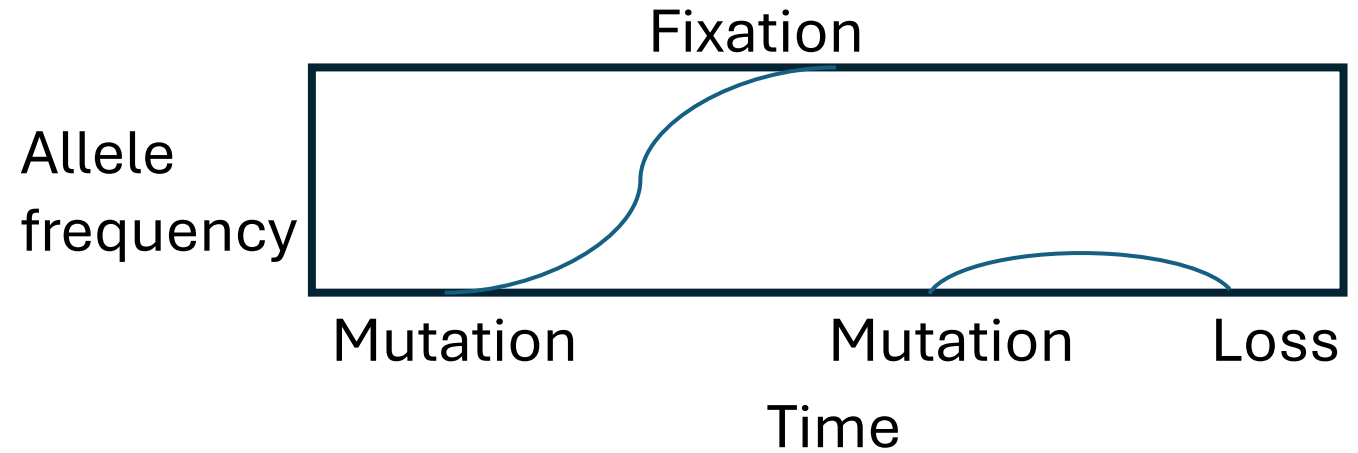


Non-synonymous substitutions

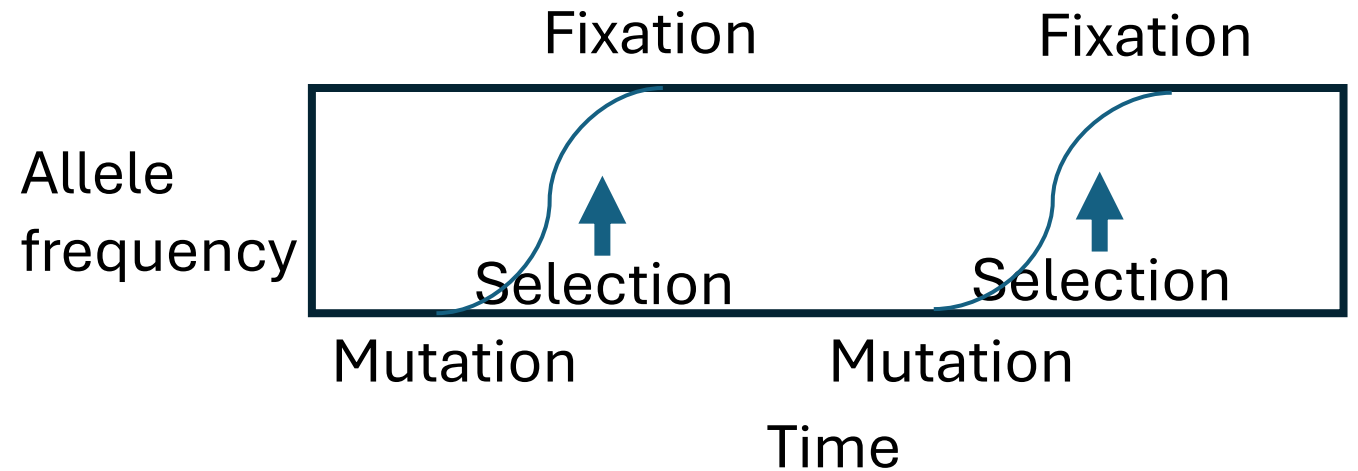


Positive selection

Synonymous substitutions



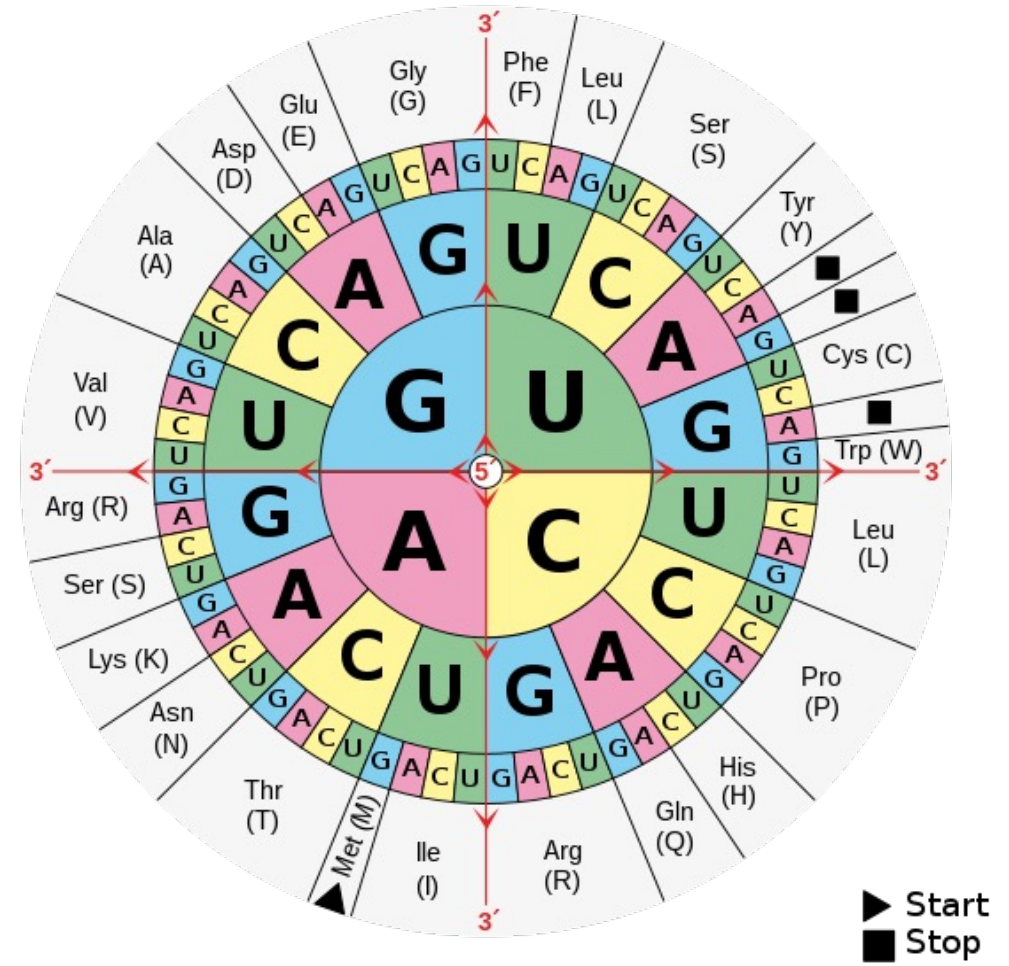
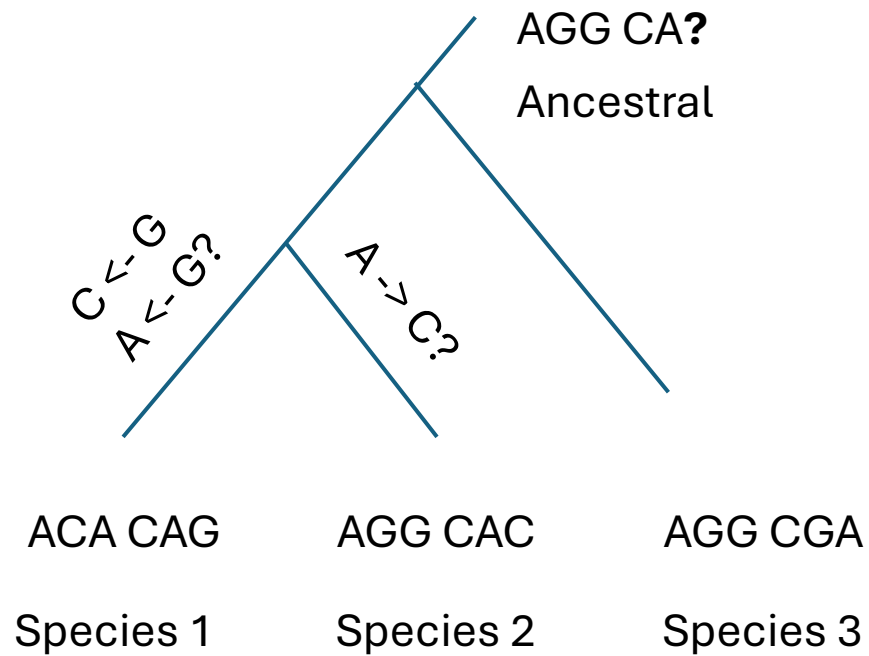
Non-synonymous substitutions



Inference of selection - d_N/d_S

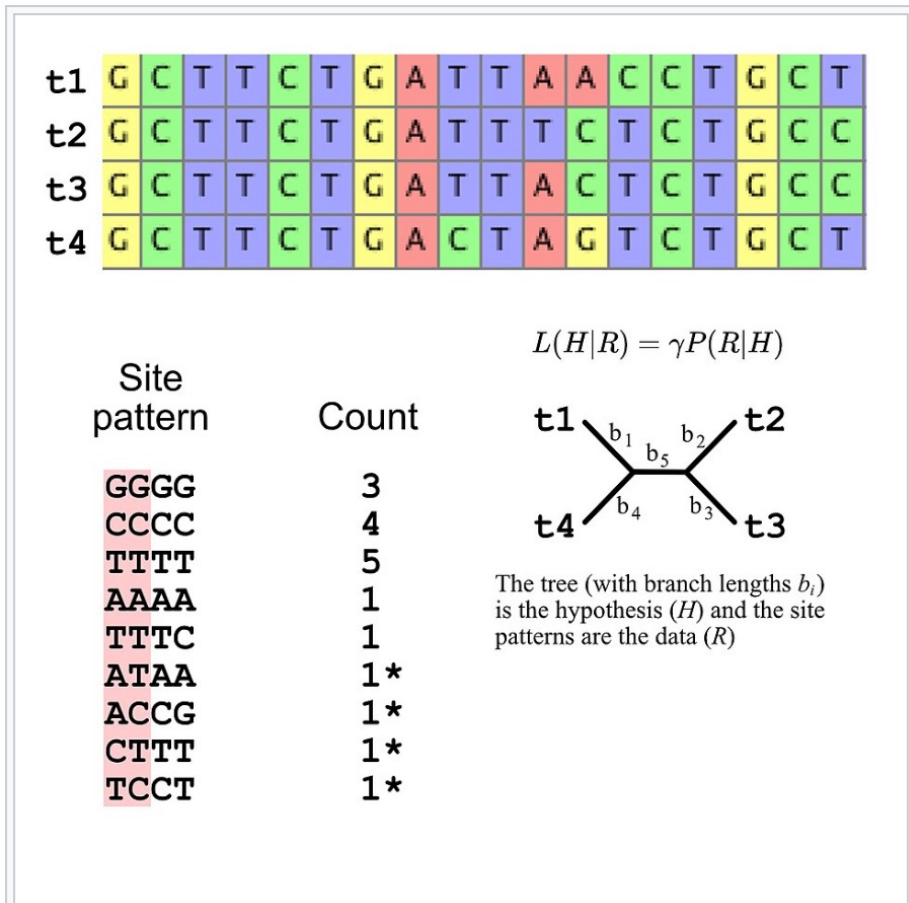
- Sequences from one individual per species can be used
 - If you have populations data – useful for excluding polymorphisms
- Rate of substitutions per synonymous site (d_S)
- Rate of substitutions per non-synonymous site (d_N)

How do we count the number of substitutions?



Infer evolutionary rate with substitution models

- Multiple sequence alignment (inferred using substitutions models)
- The data converted to a set of site patterns
- The site patterns are used to calculate the likelihood with a
 - substitution model
 - phylogenetic tree
- The substitution model is used to estimate evolutionary distances for pairs of sequence
 - Distances are the number of substitutions that have occurred since the sequences had a common ancestor
- The evolutionary distance equation (Jukes and Cantor, 1969).
 - Transforms the proportion of nucleotide differences into evolutionary distance



Substitution models

- Substitution models
 - use observed differences to try to infer actual number of substitutions during the evolutionary history

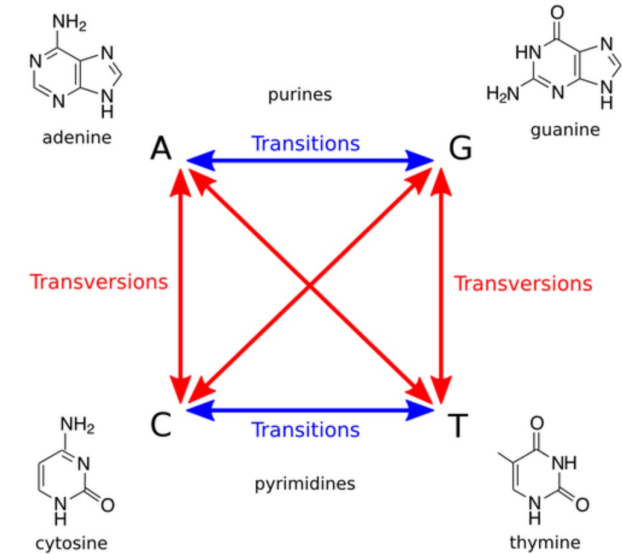
Substitution matrix

$$M_{ij} = \begin{matrix} & \begin{matrix} A & C & G & T \end{matrix} \\ \begin{pmatrix} 1-3\epsilon & \epsilon & \epsilon & \epsilon \\ \epsilon & 1-3\epsilon & \epsilon & \epsilon \\ \epsilon & \epsilon & 1-3\epsilon & \epsilon \\ \epsilon & \epsilon & \epsilon & 1-3\epsilon \end{pmatrix} & \begin{matrix} A \\ C \\ G \\ T \end{matrix} \end{matrix}$$

A-C, A-G, A-T, C-G, C-T and G-T

Substitution models, examples

JC or JC69	0	Equal substitution rates and equal base frequencies (Jukes and Cantor, 1969).
F81	3	Equal rates but unequal base freq. (Felsenstein, 1981).
K80 or K2P	1	Unequal transition/transversion rates and equal base freq. (Kimura, 1980).
HKY or HKY85	4	Unequal transition/transversion rates and unequal base freq. (Hasegawa, Kishino and Yano, 1985).
TN or TN93	5	Like HKY but unequal purine/pyrimidine rates (Tamura and Nei, 1993).
TNe	2	Like TN but equal base freq.
K81 or K3P	2	Three substitution types model and equal base freq. (Kimura, 1981).
K81u	5	Like K81 but unequal base freq.
GTR	8	General time reversal model, unequal rates and unequal base freq. (Tavare, 1986).



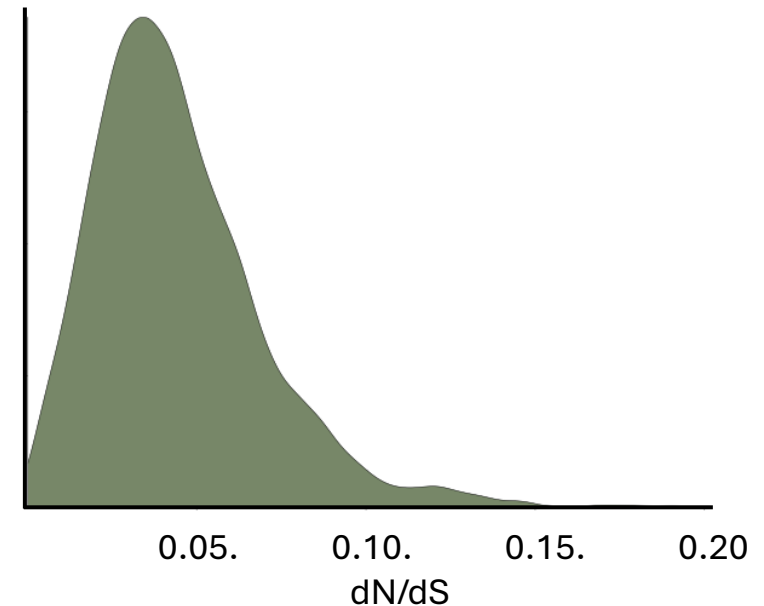
$$M_{ij} = \begin{pmatrix} 1 - 3\epsilon & \epsilon & \epsilon & \epsilon \\ \epsilon & 1 - 3\epsilon & \epsilon & \epsilon \\ \epsilon & \epsilon & 1 - 3\epsilon & \epsilon \\ \epsilon & \epsilon & \epsilon & 1 - 3\epsilon \end{pmatrix}$$

A-C, A-G, A-T, C-G, C-T and G-T

- +F base frequencies
- +I invariate sites
- +G gamma variation

Inference of selection - d_N/d_S

- Under neutrality: $d_N = d_S$
- Purifying selection: $d_N/d_S < 1$
- Positive selection: $d_N/d_S > 1$



BUSCO genes from 8 Lepidoptera lineages.
Examples of strong purifying selection.

Näsval et al. 2013

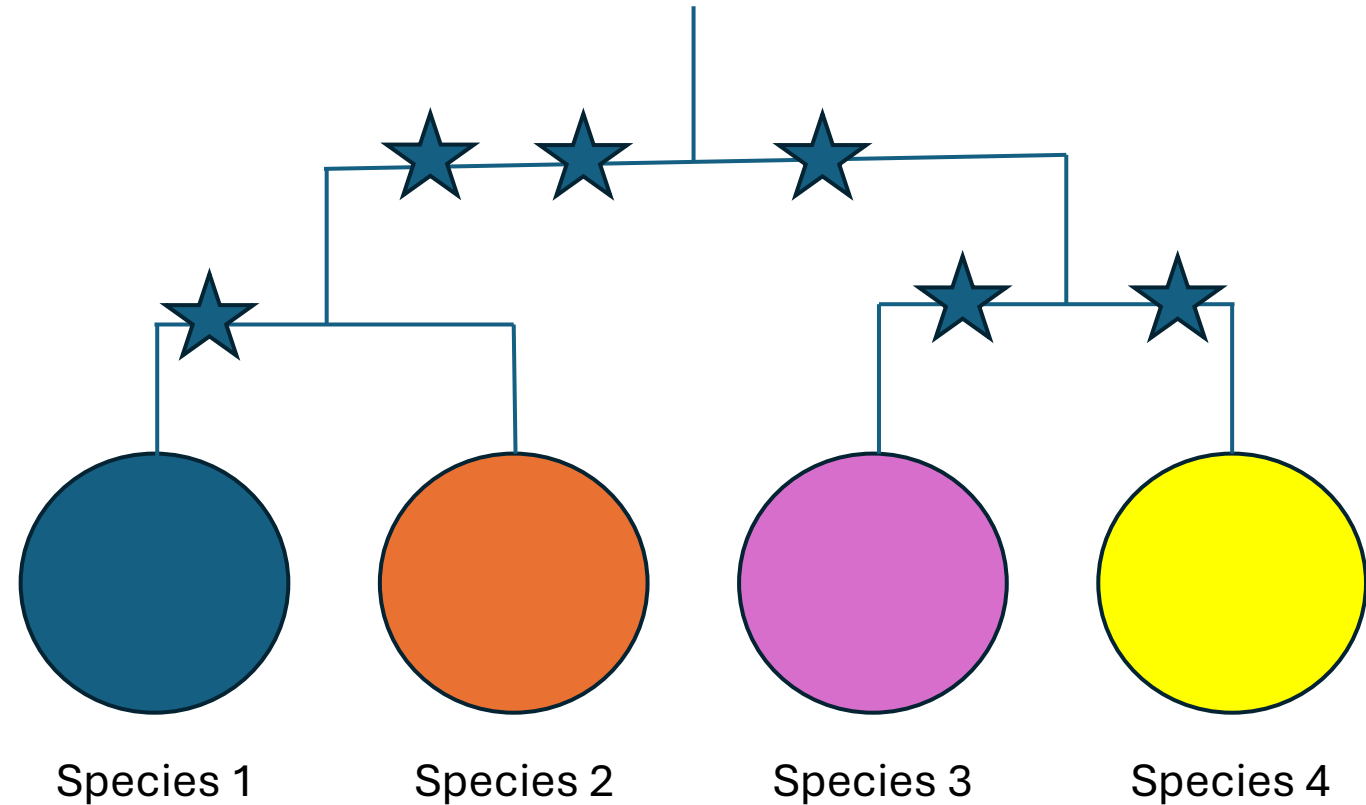
What is required?

- Codon alignment of coding nucleotide sequence
- Medium divergence
 - Total branch length > 0.5
 - Too closely related
 - Low number of fixed differences
 - Too distant
 - Substitution saturation (recurrent mutations, problem with alignment)

G	C	T	T	C	T	G	A	T	T	A	A	C	C	T	G	C	T
G	C	T	T	C	T	G	A	T	T	T	C	T	C	T	G	C	C
G	C	T	T	C	T	G	A	T	T	A	C	T	C	T	G	C	C
G	C	T	T	C	T	G	A	C	T	A	G	T	C	T	G	C	T
G	C	T	T	C	T	G	A	T	T	A	A	C	C	T	G	C	T
G	C	T	T	C	T	G	A	T	T	T	C	T	C	T	G	C	C
G	C	T	T	C	T	G	A	T	T	A	C	T	C	T	G	C	C
G	C	T	T	C	T	G	A	C	T	A	G	T	C	T	G	C	T
G	C	T	T	C	T	G	A	T	T	A	A	C	C	T	G	C	T
G	C	T	T	C	T	G	A	T	T	T	C	T	C	T	G	C	C
G	C	T	T	C	T	G	A	T	T	A	C	T	C	T	G	C	C
G	C	T	T	C	T	G	A	C	T	A	G	T	C	T	G	C	T
G	C	T	T	C	T	G	A	T	T	A	A	C	C	T	G	C	T
G	C	T	T	C	T	G	A	T	T	T	C	T	C	T	G	C	C
G	C	T	T	C	T	G	A	T	T	A	C	T	C	T	G	C	C
G	C	T	T	C	T	G	A	C	T	A	G	T	C	T	G	C	T

Tools

- codeml – model comparison detecting positive selection using codon substitution models
- mapNH – time heterogeneous models, account for non-stationary GC-content
- HyPhy, phyml, etc



codeml

Table 3
Parameters in Branch-Site Model A

Site class	Proportion of sites	Background ω	Foreground ω
0	p_0	$0 < \omega_0 < 1$	$0 < \omega_0 < 1$
1	p_1	$\omega_1 = 1$	$\omega_1 = 1$
2a	$(1 - p_0 - p_1) \cdot$ $p_0 / (p_0 + p_1)$	$0 < \omega_0 < 1$	$\omega_2 \geq 1$
2b	$(1 - p_0 - p_1) \cdot$ $p_1 / (p_0 + p_1)$	$\omega_1 = 1$	$\omega_2 \geq 1$

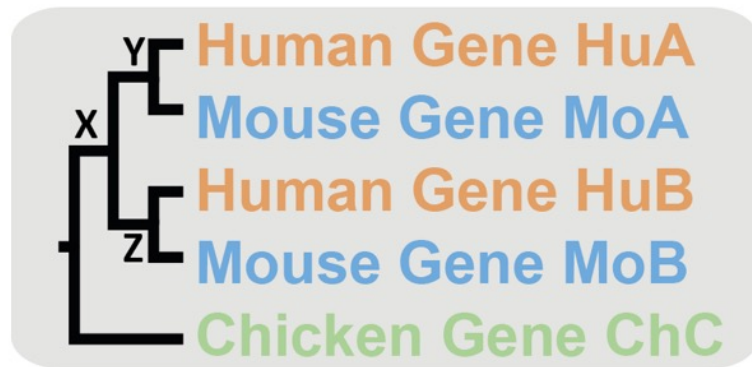
NOTE.—Model A is the alternative hypothesis for the branch-site test of positive selection. The null model fixes $\omega_2 = 1$ for the foreground branch.

Exercise

- 1) Across the genome
 - Gene annotation (we will use already prepared genesets)
 - Cluster orthogroups and select single copy orthologs – OrthoFinder (prepared)
 - Alignment - we need to align homologous sites, codon aware aligner - PRANK
 - Phylogeny reconstruction, tree building - IQ-tree
 - Infer evolutionary rates – codeml
- 2) Genes of interest
 - Use an alignment and tree from myxovirus resistance gene in vertebrates
 - Test for positive selection in one branch
 - codeml Model A – branch-site model

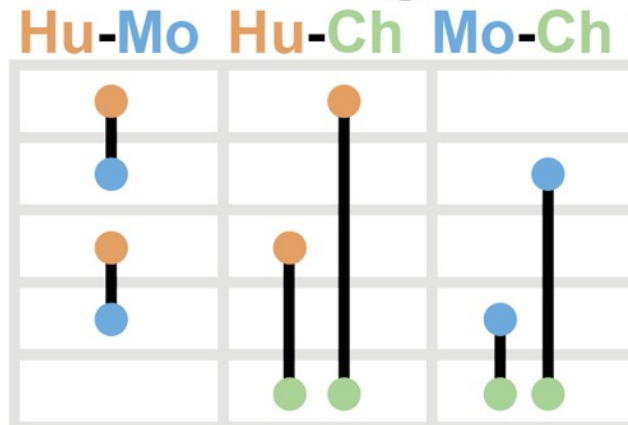
OrthoFinder

A. Orthogroup



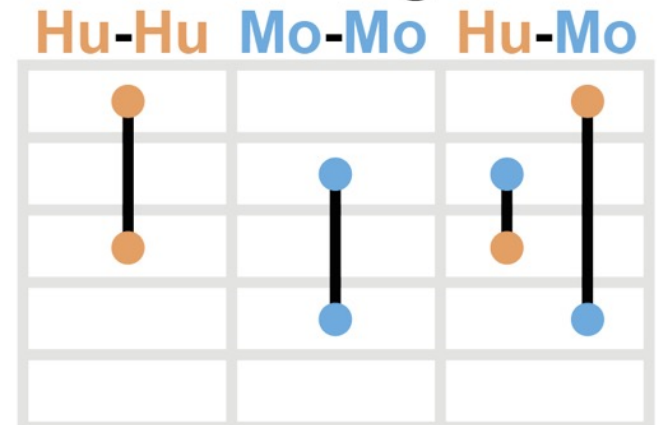
Group of genes descended from single gene in LCA of group of species

B. Orthologs



Pairs of genes descended from single gene in LCA of pair of species

C. Paralogs



Pairs of genes descended from gene duplication event

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